



EXAM PAPERS PRACTICE

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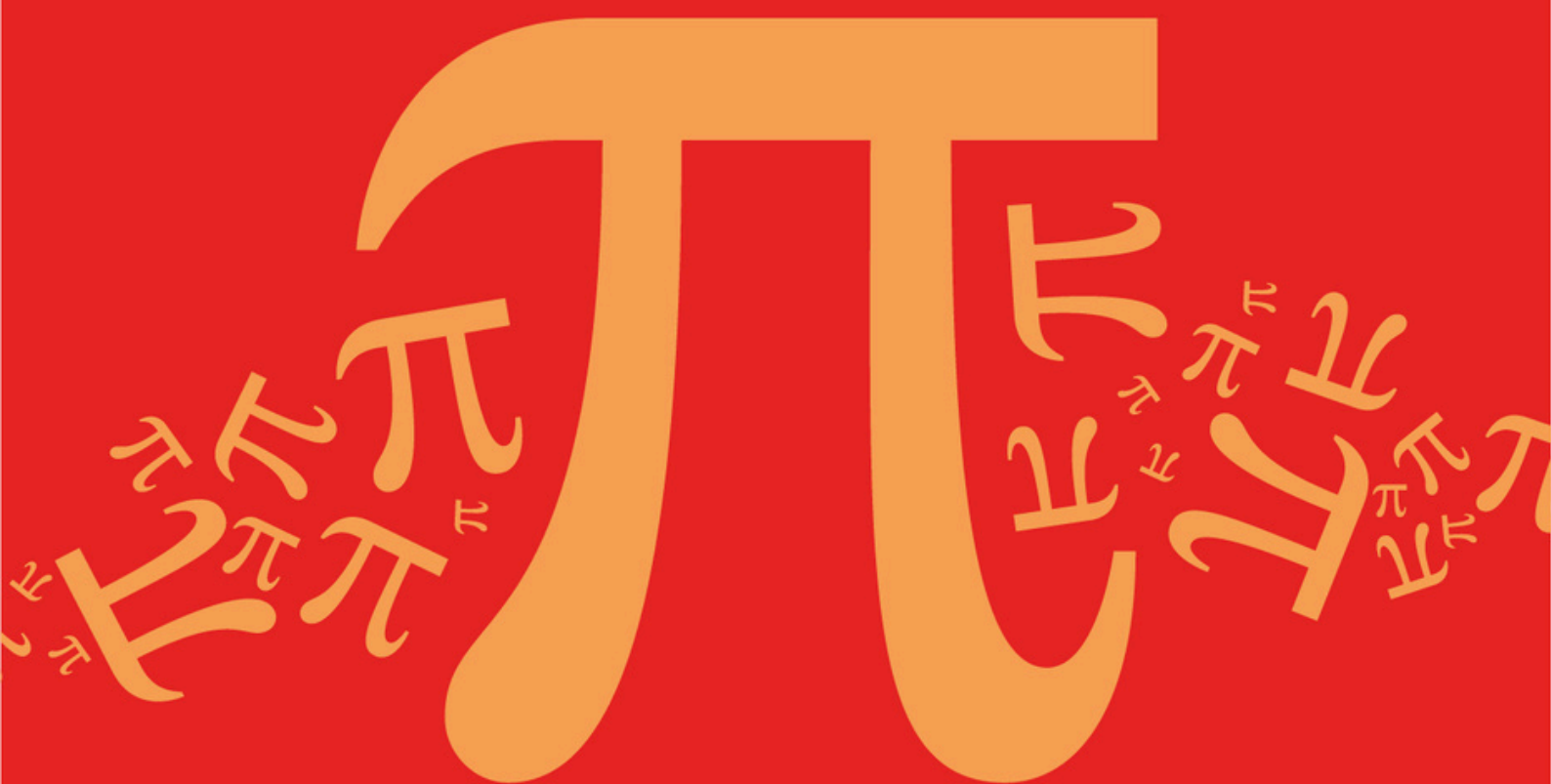
Practice questions created by actual
examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and
thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B9: Exponential Form

Mark Scheme

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B: Complex Numbers

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B9: Exponential Form

EXAM PAPERS PRACTICE

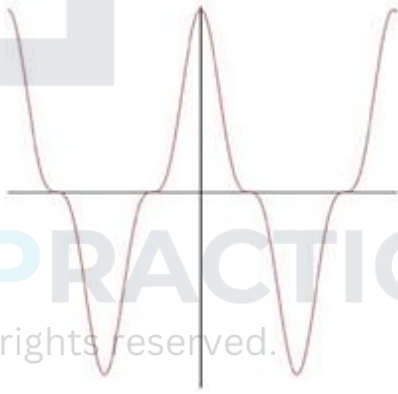
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Correctly manipulates $y = [f(x)]^n$ into a form that can be sketched.	AO3.1a	M1	$\cosh(ix) \frac{e^{ix} + e^{-ix}}{2}$
Sketches an appropriate section of the graph. Sketch does not need to be accurate but should look symmetrical about y-axis and the area above and below the x-axis should look similar	AO1.1b	M1	$= \frac{\cos(x) + t \sin(x) + \cos(-x) + t \sin(-x)}{2}$ $= \frac{\cos(x) - t \sin(x) + \cos(x) - t \sin(x)}{2}$ $= \cos(x)$
Deduces, with fully correct reasoning, the correct mean value	AO2.2a	R1	$\therefore [f(x)]^n = \cos^n(x)$  Since the graph has the same shape/area above and below the x-axis over the given domain, the mean value of $y = [f(x)]^n$ must be 0.
Total 3 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Commences an argument by correctly expanding brackets and simplifying final term to 1/16	AO1.1a	M1	$\left(1 - \frac{1}{4}e^{2i\theta}\right)\left(1 - \frac{1}{4}e^{-2i\theta}\right)$ $= 1 - \frac{1}{4}e^{2i\theta} - \frac{1}{4}e^{-2i\theta} + \frac{1}{16}$
	Substitutes correctly for	AO1.1b	B1	

	both $e^{2i\theta}$ and $e^{-2i\theta}$ in terms of $\cos 2\theta$ and $\sin 2\theta$ (seen anywhere in solution)			$= \frac{17}{16} - \frac{1}{4}(\cos 2\theta + i \sin 2\theta)$ $- \frac{1}{4}(\cos 2\theta - i \sin 2\theta)$ $= \frac{17}{16} - \frac{1}{2} \cos 2\theta$ $= \frac{1}{16}(17 - 8 \cos 2\theta)$
	Completes argument and reaches stated result by collecting terms and simplifying correctly, no errors in working seen AG	AO2.1	R1	AG
(b)	Identifies series as a geometric series and states first term and common ratio correctly	AO1.1b	B1	Geometric series with first term $r = e^{2i\theta}$ and common ratio $a = \frac{1}{4} e^{2i\theta}$
	States and uses sum to infinity formula correctly FT incorrect values for first term and common ratio	AO1.1b	B1F	$S_{\infty} = \frac{a}{1-r} = \frac{e^{2i\theta}}{1 - \frac{1}{4} e^{2i\theta}}$
(c)	Deduces that the series in part (c) is related to the real part of the series in part (b)	AO2.2a	R1	Series stated = real part of the series $e^{2i\theta} + \frac{1}{4} e^{4i\theta} + \frac{1}{16} e^{6i\theta} + \frac{1}{64} e^{8i\theta} + \dots$
	Selects an appropriate method by using the result in part (b) and multiplying appropriately to realise the denominator	AO3.1a	M1	Using result from previous part $\frac{e^{2i\theta}}{1 - \frac{1}{4} e^{2i\theta}} =$
	Substitutes to obtain an expression with cosines and sines only - using part (a) FT incorrect sum to infinity provided M1 has been awarded	AO1.1b	A1F	$\frac{e^{2i\theta}}{(1 - \frac{1}{4} e^{2i\theta})} \times \frac{(1 - \frac{1}{4} e^{-2i\theta})}{(1 - \frac{1}{4} e^{-2i\theta})}$ $= \frac{e^{2i\theta} - \frac{1}{4}}{(1 - \frac{1}{4} e^{2i\theta})(1 - \frac{1}{4} e^{-2i\theta})}$
	Identifies the real part and correctly completes the argument to reach the stated result. Only award for an error-free fully correct solution	AO2.1	R1	$\frac{\cos 2\theta - \frac{1}{4} + i \sin 2\theta}{\frac{1}{16}(17 - 8 \cos 2\theta)}$ Real part =

			$\frac{\cos 2\theta - \frac{1}{4}}{\frac{1}{16}(17 - 8\cos 2\theta)} = \frac{16\cos 2\theta - 4}{17 - 8\cos 2\theta}$
(d)	Identifies the imaginary part and states the correct expression	AO2.2a R1	Required series = imaginary part of the given series hence $\frac{\sin 2\theta}{\frac{1}{16}(17 - 8\cos 2\theta)} = \frac{16\sin 2\theta}{17 - 8\cos 2\theta}$
Total 10 marks			

Q3.

(a) $r = 8$

B1

$$\tan^{-1} \pm \frac{4\sqrt{3}}{4} \quad \text{or} \quad \pm \frac{\pi}{3} \text{ seen}$$

or $\frac{\pi}{6}$ marked as angle to Im axis with
"vector" in second quadrant on Arg diag

M1

$$\Rightarrow \theta = \frac{2\pi}{3}$$

$$-4 + 4\sqrt{3}i = 8e^{i\frac{2\pi}{3}}$$

A1

3

(b) (i) modulus of each root = 2

B1✓

use of De Moivre – dividing argument by 3

M1

$$\Rightarrow \theta = \frac{4\pi}{9}, \frac{2\pi}{9}, \frac{8\pi}{9}$$

A1 if 3 "correct" values not all in requested interval

$$2e^{-i\frac{4\pi}{9}}, 2e^{i\frac{2\pi}{9}}, 2e^{i\frac{8\pi}{9}}$$

A2

4

(ii) Area = $3 \times \frac{1}{2} \times PO \times OR \times \sin \frac{2\pi}{3}$

Correct expression for area of triangle PQR

M1

$$= 3 \times \frac{1}{2} \times 2 \times 2 \times \sin \frac{2\pi}{3}$$

correct values of lengths in formula

A1

$$= 3\sqrt{3}$$

A1cso

3

- (c) Sum of roots (of cubic) = 0
must be stated explicitly

E1

Sum of 3 roots including 1m terms
in form $r(\cos \theta + i \sin \theta)$

M1

$$2 \left(\cos \frac{(-)4\pi}{9} + \cos \frac{2\pi}{9} + \cos \frac{8\pi}{9} \right)$$

isolating real terms; correct and with "2"

A1

$$e^{-i\frac{4\pi}{9}} = \cos \frac{4\pi}{9} - i \sin \frac{4\pi}{9} \text{ seen earlier}$$

or $\cos \frac{-4\pi}{9} = \cos \frac{4\pi}{9}$ *explicitly stated to earn final A1 mark*

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0$$

AG

A1cso

4

[14]

Q4.

(a) $1, e^{\frac{2\pi i}{5}}, e^{\frac{4\pi i}{5}}, e^{\frac{-2\pi i}{5}}, e^{\frac{-4\pi i}{5}}$
accept e^θ

B1

1

(b) $\frac{z^5 - 1}{z - 1} = z^4 + z^3 + z^2 + z + 1$

B0 if assumed

B1

$$= \left(z - e^{\frac{2\pi i}{5}} \right) \left(z - e^{\frac{4\pi i}{5}} \right) \left(z - e^{\frac{-2\pi i}{5}} \right) \left(z - e^{\frac{-4\pi i}{5}} \right)$$

accept if $e^{\frac{6\pi i}{5}}$, $e^{\frac{8\pi i}{5}}$ used here

M1A1

3

(c) Correct grouping of linear factors

M1

$$e^{\frac{2\pi i}{5}} + e^{\frac{-2\pi i}{5}} = 2\cos \frac{2\pi}{5}$$

clearly shown

A1

$$\left(z^2 - 2\cos \frac{2\pi}{5} z + 1 \right) \left(z^2 - 2\cos \frac{4\pi}{5} z + 1 \right)$$

A1

$\div z^2$ to give answer

AG

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A1

4

(d) Substitute into LHS to give $w^2 + w - 1$

B1

$$\text{RHS} \left(w - 2\cos \frac{2\pi}{5} \right) \left(w - 2\cos \frac{4\pi}{5} \right)$$

B1

Solve $w^2 + w - 1 = 0$

M1

$$w = \frac{-1 \pm \sqrt{5}}{2}$$

A1

$$\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$$

A1

with reasons for choice

E1

6

[14]

Q5.

(a) (i) $4(1+i\sqrt{3}) = 8\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
 for either $4(1+i\sqrt{3})$ or $4(1-i\sqrt{3})$ used

$$= 8e^{\frac{\pi i}{3}}$$

If either r or θ is incorrect but the same value in both
 (i) and (ii) allow A1 but for θ only if it is given as $\frac{\pi}{6}$

M1

A1

(ii) $4(1-i\sqrt{3}) = 8e^{\frac{-\pi i}{3}}$

A1

3

(b) $z^3 - 4 = \pm\sqrt{-48}$
 taking square root

M1

$$z^3 = 4 \pm 4\sqrt{3}i$$

AG

A1

2

(c) (i) $z = 2e^{\frac{\frac{\pi}{3} + 2k\pi i}{3}}$ or $z = 2e^{\frac{\frac{-\pi}{3} + 2k\pi i}{3}}$
 for the 2; ft incorrect 8, but no decimals

B1F

for either, PI

M1

$$z = 2e^{\frac{j}{9}}, 2e^{\frac{7j}{9}}, 2e^{\frac{5j}{9}}$$

$$2e^{\frac{-j}{9}}, 2e^{\frac{-7j}{9}}, 2e^{\frac{-5j}{9}}$$

Allow A1 for any 2 roots not +/- each other

Allow A2 for any 3 roots not +/- each other

Allow A3 for all 6 correct roots

Deduct A1 for each incorrect root in the interval;

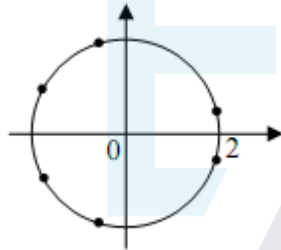
ignore roots outside the interval

ft incorrect r

A3,2,1F

5

(ii)



Radius 2

clearly indicated; ft incorrect r

allow B1 for 3 correct points condone lines

B1F

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Plotting roots

B2,1

3

- (d) (i) Sum of roots = 0 as coefficient of $z^5 = 0$
OE

E1

1

(ii)

Use of, say, $\frac{1}{2} \left(e^{\frac{j}{9}} + e^{\frac{-j}{9}} \right) = \cos \frac{\pi}{9}$

M1

$$\cos \frac{3\pi}{9} = \frac{1}{2} \text{ used}$$

A1

$$\cos \frac{\pi}{9} + \cos \frac{3\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{7\pi}{9} = \frac{1}{2}$$

AG

A1

3

[17]

Q6.

(a) (i) $\beta = 2 - 2\sqrt{3}i$

B1

1

(ii) $\alpha\beta\gamma = -8$
 Allow for +8 but not ± 16

$$\alpha\beta = 16$$

M1

B1

$$\gamma = -\frac{1}{2}$$

A1

3

(iii) Either $\frac{-p}{2} = \alpha + \beta + \gamma$

or $\frac{q}{2} = \alpha\beta + \beta\gamma + \gamma\alpha$

SC if failure to divide by 2 throughout,
 allow M1A1 for either p or q correct ft

M1

$$p = -7, \quad q = 28$$

ft incorrect γ

A1F,A1F

3

Alternative to (a)(ii) and (a)(iii)

$$(z^2 - 4z + 16)(az + b)$$

(M1)

$$\alpha\beta = 16$$

(B1)

$$a = 2, b = +1, \gamma = -\frac{1}{2}$$

(A1)

Equating coefficients

(M1)

$$p = -7$$

(A1F)

$$q = 28$$

(A1F)

(b) (i) $r = 4, \theta = \frac{\pi}{3}$

B1,B1

2

(ii) $(2 + 2\sqrt{3}i)^n = \left(4e^{i\frac{\pi}{3}}\right)^n$

M1

$$= 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

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AG

A1

2

(iii) $(2 - 2\sqrt{3}i)^n = 4^n \left(\cos \frac{n\pi}{3} - i \sin \frac{n\pi}{3} \right)$

B1

$$\begin{aligned} \alpha^n + \beta^n + \gamma^n &= 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right) \\ &+ 4^n \left(\cos \frac{n\pi}{3} - i \sin \frac{n\pi}{3} \right) + \left(-\frac{1}{2} \right)^n \end{aligned}$$

M1

$$= 2^{2n+1} \cos \frac{n\pi}{3} + \left(-\frac{1}{2} \right)^n$$

AG

A1

3

[14]

Q7.

(a) (i) $\left(e^{\frac{2\pi i}{7}}\right)^7 = e^{2\pi i} = 1$

Or $z^7 = e^{2k\pi i} \quad z = e^{\frac{2k\pi i}{7}} \quad k = 1$

B1

1

(ii) Roots are $\omega^2, \omega^3, \omega^4, \omega^5, \omega^6$

OE; M1A0 for incomplete set

SC B1 for a set of correct roots in terms of $e^{i\theta}$

M1A1

2

(b) Sum of roots considered

M1

= 0

Or $\sum_{r=0}^6 \omega^r = \frac{\omega^7 - 1}{\omega - 1} = 0$

A1

2

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(c) (i) $\omega^2 + \omega^5 = e^{\frac{4\pi i}{7}} + e^{\frac{10\pi i}{7}}$

M1

$= e^{\frac{4\pi i}{7}} + e^{\frac{-4\pi i}{7}}$

Or $\cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} + \cos \frac{4\pi}{7} - i \sin \frac{4\pi}{7}$

A1

$= 2 \cos \frac{4\pi}{7}$

AG

A1

3

(ii) $\omega + \omega^6 = 2 \cos \frac{2\pi}{7}; \quad \omega^3 + \omega^4 = 2 \cos \frac{6\pi}{7}$

Allow these marks if seen earlier in the solution

B1,B1

Using part (b)

M1

Result

AG

A1

4

[12]

Q8.

(a) (i) $1 + \sqrt{3}i = 2e^{\frac{\pi i}{3}}$
B1 both correct

B1

$1 - i = \sqrt{2}e^{-\frac{\pi i}{4}}$
OE

B1B1

3

(ii) $2^{\frac{21}{2}}$ or equivalent single expression
No decimals; must include fractional powers

B1F

Raising and adding powers of e

M1

$\frac{17\pi}{12}$ or equivalent angle
Denominators of angles must be different

A1F

3

(b) $z = \sqrt[3]{2^{10}\sqrt{2}} e^{\frac{17\pi i}{36} + \frac{2\pi i}{3}}$

M1

$\sqrt[3]{2^{10}\sqrt{2}} = 8\sqrt{2}$
CAO

B1

$$\theta = \frac{17\pi}{36}, -\frac{7\pi}{36}, -\frac{31\pi}{36}$$

Correct answers outside range: deduct 1 mark only

A2,1F

4

[10]

Q9.

- (a) Correct multiplication of brackets

M1

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta$$

Clearly shown

A1

2

- (b) $2 \cos \theta = 1$

SC If 'hence' not used and, say,
 $z^8 - z^4 + 1 = 0$ is solved by formula, lose

M1

$$\theta = \frac{\pi}{3}$$

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 M1A1, but then continue M1m1 etc if $\frac{\pi}{3}$ is obtained

A1

$$z^4 = e^{\frac{\pi i}{3}} \text{ or } e^{-\frac{\pi i}{3}}$$

M1

$$z = e^{\frac{\pi i}{12} + \frac{2k\pi i}{4}} \text{ or } e^{-\frac{\pi i}{12} + \frac{2k\pi i}{4}}$$

m1

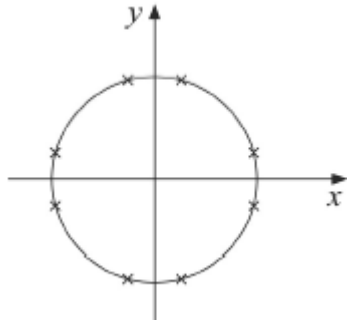
$$e^{\pm \frac{\pi i}{12}}, e^{\pm \frac{7\pi i}{12}}, e^{\pm \frac{5\pi i}{12}}, e^{\pm \frac{11\pi i}{12}}$$

A1 if 3 roots correct

A2,1,0F

6

- (c)



B1 for 4 roots indicated correctly on a circle.
CAO

B2,1,0

Indication that $r = 1$

B1

3

[11]

Q10.

(a) $z^4 = 16e^{\frac{4\pi i}{12}}$

Allow M1 if $z^4 = 2e^{\frac{4\pi i}{12}}$

M1

$$= 16 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

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OE could be $2ae^{\frac{\pi i}{3}}$ or

$$2a \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

A1

$$= 8 + 8\sqrt{3}i, \alpha = 8$$

ft errors in 2^4

A1F

3

(b) For other roots, $r = 2$

for realising roots are of form $2 \times e^{i\theta}$.

M1 for strictly correct θ

B1

$$\theta = \frac{\pi}{12} + \frac{2k\pi}{4}$$

i.e must be $\left(\text{their } \frac{\pi}{3} + 2k\pi \right) \times \frac{1}{4}$

M1A1

Roots are $2e^{\frac{7\pi i}{12}}, 2e^{\frac{-5\pi i}{12}}, 2e^{\frac{-11\pi i}{12}}$

ft error in $\frac{\pi}{12}$ or r

$\left[\text{accept } 2e^{\left(\frac{\pi}{12} + \frac{2k\pi}{4} \right)i} \quad k = -1, -2, 1 \right]$

A2,1,0F

5

[8]

Q11.

- (a) Any method for finding r or θ

M1

$r = 4\sqrt{2}, \theta = \frac{\pi}{4}$

A1A1

3

- (b) $z^5 = 4\sqrt{2} e^{\frac{\pi i}{4}}$ 2025 Exam Papers Practice. All rights reserved.

M1 needs some reference to $a + 2k\pi i$

$z = \sqrt{2} e^{\frac{\pi i}{20} + \frac{2k\pi i}{5}}$

A1 for r
A1 for θ] incorrect r, θ part (a)

M1A1FA1F

$z = \sqrt{2} e^{\frac{\pi i}{20}}, \sqrt{2} e^{\frac{9\pi i}{20}}, \sqrt{2} e^{\frac{17\pi i}{20}}, \sqrt{2} e^{\frac{-7\pi i}{20}}, \sqrt{2} e^{\frac{-15\pi i}{20}}$

Accept r in any form eg $32^{\frac{1}{10}}$

Correct but some answers outside range
allow A1

ft incorrect r, θ in part (a)

A2,1,0F

5

[8]

Q12.

(a) $1, e^{\pm \frac{2\pi i}{3}}$

M1 for any method which would lead to the correct answers
Accept e^0 or e^{0i}
Also accept answers written down correctly

M1A1

2

(b) Any correct method

M1

Shown for one root

AG

A1

2

(c) (i) $\frac{\omega}{\omega+1} = \frac{\omega}{-\omega^2}$

ie use of result in (b)

$= -\frac{1}{\omega}$

AG

A1

2

(ii) $\frac{\omega^2}{\omega^2+1} = -\omega$

AG

A1

1

(iii) $\left(\frac{\omega}{\omega+1}\right)^k + \left(\frac{\omega^2}{\omega^2+1}\right)^k = \left(-\frac{1}{\omega}\right)^k + (-\omega)^k$

M1A1

Use of $\omega = e^{\frac{2\pi i}{3}}$

m1

$= (-1)^k \left(e^{\frac{-2k\pi i}{3}} + e^{\frac{2k\pi i}{3}} \right)$

A1

$$= (-1)^k 2 \cos \frac{2k\pi}{3}$$

AG

A1

5

[12]

Q13.

(a) (i) $z^3 = \frac{4 \pm \sqrt{16 - 32}}{2}$

M1

$= 2 \pm 2i$
AG

A1

2

(ii) $2 + 2i = 2\sqrt{2}e^{\frac{\pi i}{4}}, 2 - 2i = 2\sqrt{2}e^{\frac{-\pi i}{4}}$

M1 for either result or for one of $r = 2\sqrt{2}, \theta = \pm \frac{\pi}{4}$

M1

$\left(r = 2\sqrt{2} \quad A1, \theta = \pm \frac{\pi}{4} \quad A1 \right)$

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A1A1

$$z = \sqrt{2}e^{\frac{\pi i}{12} + \frac{2k\pi i}{3}} \text{ or } \sqrt{2}e^{\frac{-\pi i}{12} + \frac{2k\pi i}{3}}$$

M1 for either

M1

$$z = \sqrt{2}e^{\frac{\pm \pi i}{12}}, \sqrt{2}e^{\frac{\pm 3\pi i}{4}}, \sqrt{2}e^{\frac{\pm 7\pi i}{12}}$$

allow A1 for any 3 correct ft errors in $\pm \frac{\pi}{4}$

A2,1,0 F

6

(b) Multiplication of brackets

M1

Use of $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$

AG

A1

2

$$(c) \quad \left(z - \sqrt{2}e^{\frac{\pi i}{12}} \right) \left(z - \sqrt{2}e^{-\frac{\pi i}{12}} \right)$$

$$= z^2 - 2\sqrt{2} \cos \frac{\pi}{12} z + 2$$

PI

M1A1F

$$\left(z^2 - 2\sqrt{2} \cos \frac{\pi}{12} z + 2 \right)$$

Product is $\left(z^2 - 2\sqrt{2} \cos \frac{7\pi}{12} z + 2 \right)$
 (or $z^2 + 2z + 2$)

A1F

3

$$(z^2 - 2\sqrt{2} \cos \frac{3\pi}{4} z + 2)$$

[13]

Q14.

$$(a) \quad \frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = i$$

AG

M1A1

2

$$(b) \quad |z_2| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1 = |z_1|$$

M1A1

2

$$(c) \quad r = 1$$

PI

B1

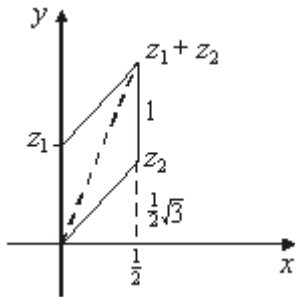
$$\theta = \frac{1}{2}\pi, \frac{1}{3}\pi$$

Deduct 1 mark if extra solutions

B1B1

3

(d)



Positions of the 3 points, relative to each other, must be approximately correct

B2,1F

2

(e) $\text{Arg}(z_1+z_2) = \frac{5}{12}\pi$

Clearly shown

B1

$$\tan \frac{5}{12}\pi = \frac{1 + \frac{1}{2}\sqrt{3}}{\frac{1}{2}}$$

Allow if B0 earned

M1

$$= 2 + \sqrt{3}$$

AG must earn B0 for this

B1

3

[12]

Q15.

(a) $z = e^{\frac{2k\pi i}{6}}, \quad k = 0, \pm 1, \pm 2, 3$

M1

OE

M1A1 only if:

(1) range for k is incorrect e.g. $0, 1, 2, 3, 4, 5$

(2) i is missing

A2,1,0

3

(b) (i) $\frac{w^2-1}{w} = w - \frac{1}{w} = 2i \sin \theta$

AG

M1A1

2

$$(ii) \quad \frac{w}{w^2 - 1} = \frac{1}{2i \sin \theta}$$

M1

$$= -\frac{i}{2 \sin \theta}$$

AG

A1

2

$$(iii) \quad \frac{2i}{w^2 - 1} = \frac{-2iw^{-1}i}{2 \sin \theta}$$

$$\text{Or for } \frac{1}{\sin \theta e^{i\theta}}$$

M1

$$= \frac{1}{\sin \theta} (\cos \theta - i \sin \theta)$$

A1

$$= \cot \theta - i$$

AG

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A1

3

$$(iv) \quad z = \frac{2i}{w^2 - 1} \text{ Or } z + 2i = \frac{2i}{w^2 - 1} + 2i$$

ie any correct method

M1

$$z + 2i = zw^2$$

AG

A1

2

(c) (i) No coefficient of z^6

E1

1

$$(ii) \quad (w^2)^6 = 1 \quad w^2 = e^{\frac{k\pi i}{3}}$$

B1

$$z = \cot \frac{k\pi}{6} - i, \quad k = \pm 1, \pm 2, 3$$

M1
A2, 1, 0

Alternatively:

$$z + 2i = e^{\frac{k\pi i}{3}} z$$

B1

$$z = \frac{2i}{e^{\frac{k\pi i}{3}} - 1}$$

M1

roots A2, 1, 0

(NB roots are $\pm \sqrt{3} - i$; $\pm \frac{1}{\sqrt{3}} - i$; $-i$)

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Examiner reports

Q3.

- (a) Many students seemed unable to find the correct argument of the complex number. Students would be well advised to make a small sketch of the Argand diagram in order to determine the correct quadrant for the angle θ .
- (b) (i) An incorrect value of θ in part (a) prevented a large number of students from obtaining the correct roots of the cubic equation. Most students had the correct modulus and realised the need to divide an argument by 3, but did not always apply de Moivre's Theorem correctly.
- (ii) It was possible for students to score full marks here if they had the correct modulus of the 3 roots, provided they recognised that the points P , Q and R were the vertices of an equilateral triangle. Poor trigonometrical skills prevented most students from obtaining the correct value for the area of the triangle.
- (c) It was necessary to state explicitly that the sum of the roots of the cubic equation was zero. By considering the sum of the roots and then taking real parts of both sides of the equation the printed result followed. Better students stated explicitly that
- $$\cos\left(-\frac{4\pi}{9}\right) = \cos\left(\frac{4\pi}{9}\right)$$
- at some stage in their proof.

Q4.

Responses to this question were extremely poor and very few candidates were able to make any headway at all after part (a). Very few candidates seemed to understand what linear factors were – most merely wrote down the roots of z^5 in some other form. This in turn meant that virtually no one was able to answer part (c). There were not many serious attempts at part (d) in spite of the fact that it could be done by assuming the result in part (c), although some candidates were able to pick up the odd mark.

Q5.

Parts (a) and (b) of this question were well done; part (c)(i) was reasonably well done especially by the more able candidates. However, the Argand diagrams were very poorly drawn. Many candidates drew a rough circle by hand, with no indication of its radius, and then went on to put crosses on their circle at points which bore little resemblance to the angles in question. Some candidates drew no circle at all but drew lines of differing lengths from the origin. There were few correct responses to part (d) as candidates failed to realise the relevance of part (c), and especially part (d)(i), to this last part.

Q6.

This question provided a good source of marks for many candidates. The commonest error in part(a) was to write $\alpha\beta\gamma$ as -16 , overlooking the fact that the coefficient of x^3 was not unity and, of course, leading to an incorrect value for γ . This error perpetuated in part (a)(ii) with $\alpha + \beta + \gamma$ written as p or $-p$ and the same for q .

Parts (b)(i) and (b)(ii) were generally well done, although it should be stated that when answers are printed, candidates are expected to provide sufficient detail to show clearly how their answers are arrived at. Part (b)(iii) was also quite well done and it was pleasing

to note that in some cases where candidates had not arrived at $\gamma = -\frac{1}{2}$, they went back to part (a)(ii) to identify their error. Just occasionally in part (b)(iii), some candidates made

blatant errors in their attempt to convert $4^x \cos \frac{22\pi}{3} + 4^x \cos \frac{22\pi}{3}$ into $2^{2x+1} \cos \frac{22\pi}{3}$.

Q7.

Whilst part (a) was generally well done, relatively few candidates expressed the other roots in terms of ω , but rather gave them in the form $re^{i\theta}$.

Few, also, were able to complete part (b). The relation $1 + \omega + \omega^2 = 0$ appeared with regularity.

Part (c) was poorly answered. Although correct answers were written down as they were given in the question, few responses were convincing and as has already been stated earlier, if answers are given, it is the responsibility of the candidates to supply sufficient working to convince the examiner that they understand the methods involved.

Q8.

Part (a)(i) was generally well done. The less successful candidates usually wrote the argument of $1 - i$ as $3\pi/4$ instead of $-\pi/4$. In part (a)(ii) there was some poor handling of fractions in the argument of the product of the two complex numbers, and also some omission of raising the moduli of the two complex numbers to their respective powers. Many of the candidates who had been successful in part (a) often went on to complete part (b) correctly, although some candidates lost marks either through not giving z in the form asked for or by giving values for θ outside the specified range.

Q9.

Part (a) was quite well done, although for some reason the multiplication of the brackets sometimes resulted in $z^8 - 2(e^{i\theta} + e^{-i\theta})z^4 + 1$ followed by the printed result. In part (b), a number of candidates lost some marks through attempting to use an 'otherwise' method

instead of the 'hence' method as directed. Those candidates arriving at $z^4 = e^{\frac{\pi i}{3}}$ usually went on to solve the given equation correctly. The Argand diagram, however, was poorly drawn. Frequently no circle was indicated and roots appeared at different distances from the origin, and in many cases candidates seemed to think that the eight roots were equally spaced round the origin.

Q10.

This question proved to be slightly more demanding for candidates than had been

anticipated. The main difficulty in part (a) was that, having written down $2^4 e^{\frac{\pi i}{3}}$ (or

frequently incorrectly $2e^{\frac{\pi i}{3}}$), candidates were unsure about how to proceed, and they either abandoned this part of the question at that point or then tried to manipulate $a(1 + \sqrt{3}i)$ with little success.

In part (b), candidates were divided into two clear categories: those who had been clearly taught how to find roots of equations in the form $re^{i\theta}$ and those who were very muddled in

their thinking. It generally hinged on candidates' ability to write θ as $\frac{\pi}{12} + \frac{k\pi}{2}$. Quite a number of otherwise correct solutions lost a mark through giving values of θ outside the specified range.

Q11.

Part (a) was done well by the majority of candidates. However, responses to part (b) were less successful. A number of candidates gave the roots of $z^5 = 1$ as their answer to part (b), whilst others left the modulus of the roots as $4\sqrt{2}$ instead of $\sqrt{2}$, and others again gave solutions outside the range of θ as specified in the question. Some candidates yet

again were either unable to handle $\frac{1}{5} \left(\frac{\pi}{4} + 2k\pi \right) i$ or, when taking the fifth root of $e^{\left(\frac{\pi}{4} + 2k\pi \right) i}$, wrote $e^{\left(\frac{\pi}{4} + \frac{2k\pi}{5} \right) i}$.

Q12.

It was disappointing to find many candidates unsure of the cube roots of unity and even more unsure of how to obtain them. It was also disappointing to note that few candidates were able to establish the result $1 + \omega + \omega^2 = 0$ in part (b), in spite of the variety of ways in which this result could be established. On the whole, parts (c)(i) and (c)(ii) were correctly done in spite of using roundabout methods to obtain the printed results. In part (c)(iii),

however, most solutions ended at $\left(-\frac{1}{\omega} \right)^k + (-\omega)^k$ but of those candidates who attempted this part further, sign errors hindered completely correct solutions.

Q13.

Candidates were usually able to establish the result in part (a) although the methods used were sometimes somewhat inelegant. Part (a)(ii) was reasonably well done although some carelessness was in evidence in this part. For instance, some candidates although

showing that the argument of z^3 was $\pm \frac{1}{4} \pi$ continued their solution with only $\frac{1}{4} \pi$ and so arrived at a total of three roots. Others having reached $|z^3| = \sqrt{8}$ then thought that $|z| = \sqrt{8}$ also.

A few candidates used a method which, although possible, was not really suitable. They replaced the z^3 in $z^6 - 4z^3 + 8 = 0$ with $2 \pm 2i$ and so arrived at $z^6 = \pm 8i$. This latter equation gave the twelve roots of $z^{12} = -64$ and the method was incomplete unless 6 of the roots were rejected. Part (b) was generally well done, but part (c) was really only completed by candidates who had correctly answered part (a)(ii).

Q14.

Part (a) was well done, as was part (b). In part (c) it was surprising to note how many candidates could not express the complex number in the form $re^{i\theta}$, although z_2 was almost invariably correctly written as $e^{i\frac{\pi}{3}}$. Errors in part (c) however did not deter candidates from

drawing a correct Argand diagram as they usually used the form $a + ib$ when plotting their points. Although the vast majority of scripts ended with $\tan \frac{5\pi}{12} = 2 + \sqrt{3}$, very, very few candidates gave convincing proof that $\arg(z_1 + z_2)$ was $\frac{5\pi}{12}$, but rather seemed to take it for granted.

Q15.

Although part (a) of this question was standard work it was surprising to see many candidates fail to obtain full marks. The commonest errors were either to express the six roots of $z^6 = 1$ in the form $a + i$, or to give the roots in the range 0 to $27i$. A few candidates wrote down the 6 roots as $e^{\frac{k\pi i}{3}}$ with $k = \pm 1, \pm 2, \pm 3$. In part (b), parts (i) and (iv) were often well done, but relatively few candidates spotted w part (b)(ii) as the reciprocal of part

(b)(i), and it was not unusual to see $\frac{w}{w^2 - 1}$ rewritten as $w^{-1} - w$.

Part (b)(iii) was beyond all but the most able candidates although quite a number arrived at $\frac{1}{\sin \theta e^{i\theta}}$ at which point their solutions usually petered out. There was a wide variety of reasons why the equation $(z + 2i) = z^6$ had only 5 roots with about 50% of them spurious. In part (c)(ii) only one or two candidates used the hints given in the earlier parts of the question, but instead, solved the equation $(z + 2i)^6 = z^6$ from first principles by writing $z + 2i$

$$= ze^{\frac{k\pi i}{3}} \text{ followed by } z = \frac{2i}{e^{\frac{k\pi i}{3}} - 1}.$$

Of the few serious attempts made by candidates at this part of the question, most solutions ended at the point indicated and only the most able candidates found the five roots of the equation in the required form.